

Shantanu Shastri

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EDUCATION

Carnegie Mellon University	December 2026
Master of Information Systems Management	GPA: 3.85
Manipal Institute of Technology	August 2022
Bachelor of Electronics and Communications Engineering	GPA: 3.99
<i>Relevant Coursework:</i> Machine Learning, Distributed Systems, Data Structures, Algorithms, Advanced Databases	

SKILLS

Languages: Scala, Python, Java, C++, TypeScript, JavaScript, SQL, Bash/Shell
ML Infrastructure: Training Datasets, Point-in-Time Joins, Feature Engineering, Attribution Data
Big Data & Pipelines: Apache Spark, Ray Data, Apache Iceberg, Dataflow Pipelines, Apache Kafka
Systems & Generative AI: Generative AI Integration, Docker, Kubernetes, CI/CD, Storage Format Optimization

EXPERIENCE

BlackRock	January 2022 – August 2026
<i>AI Engineer Internship</i>	June 2026 – August 2026
Data & ML Infrastructure: Built scalable Data/ML infrastructure in Scala and Python to help applied scientists generate complex training datasets, enabling rapid and efficient ML model development. Generative AI & Datasets: Owned platforms for point-in-time joins, integrating generative AI with a hybrid vector-graph pipeline (Milvus, Neo4j) to securely generate data across 3.5M records. Spark & Cost Optimization: Contributed to large-scale Apache Spark and Ray Data pipelines by optimizing storage formats, successfully reducing runtime and computing cost while analyzing \$14T in cash.	
<i>Software Developer 2 - Associate</i>	January 2025 – July 2025
Dataflow Pipelines: Engineered scalable dataflow pipelines and Spring Boot data frameworks from scratch, managing complex data schemas to accurately automate 1.5M+ client touchpoints. Machine Learning & Systems: Optimized read-through caching layers to efficiently serve high-throughput models, applying systems engineering knowledge to cut query formulation time from 450ms to 150ms.	
<i>Software Developer 1 - Analyst</i>	July 2022 – December 2024
Code Reviews & CI/CD: Wrote clear and well-tested code while participating in rigorous code reviews, building comprehensive Azure CI/CD pipelines to ensure accurate, reliable infrastructure. Storage Format Optimization: Spearheaded the optimization of legacy storage formats, refactoring PostgreSQL schemas on tables exceeding 50M+ rows to achieve a 37% reduction in query latency. Cross-Functional Collaboration: Worked cross-functionally with data teams and applied scientists to clarify requirements, engineering a sharded server environment that boosted throughput to 10K+ RPS.	
Shunya	June 2020 – September 2020
<i>Data Science Intern</i>	
ML & Systems Curiosity: Demonstrated deep interest in machine learning and systems by engineering GPU-accelerated PyTorch workflows on Linux to automate complex computer vision evaluations. Generative AI Tooling: Used ML and generative AI tools thoughtfully to develop clustering models on multidimensional assessment data, establishing a growth mindset to dynamically recommend practice.	

PROJECTS

High-Performance Modular Booking Architecture
Scalable Infrastructure: Engineered large-scale ML infrastructure using Apache Kafka and REST APIs, seamlessly optimizing data storage and handling bursts of 100,000+ real-time requests.
Zulip Open-Source Engine Extensions
Attribution Data Generation: Built a high-performance message engine in Node.js and Redis to reliably manage attribution data streams, focusing on accurate, highly efficient data processing at scale.